

Calculus II

MTH 231 | Section C | Fall 2012

Dr. James Ashe

Johnson C. Smith

beamer-ics

Calc II: 4.8 Antiderivatives

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Using limits to avoid division by zero

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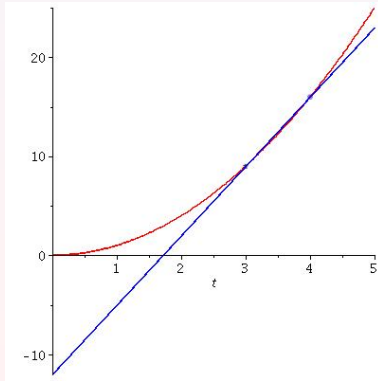
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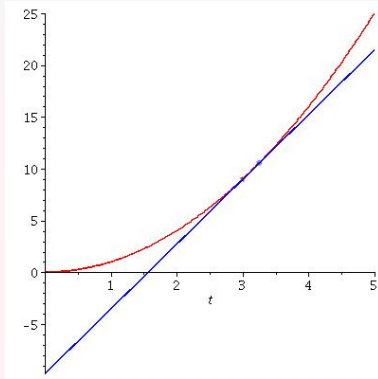
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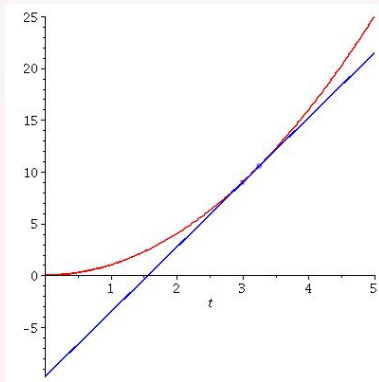
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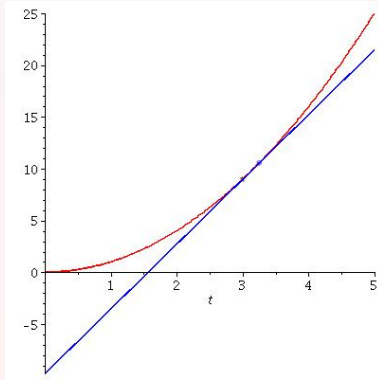
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Review

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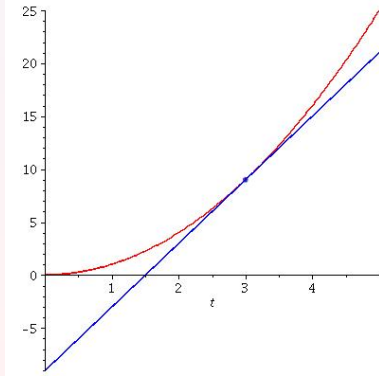
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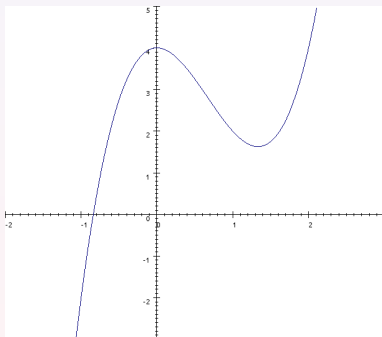
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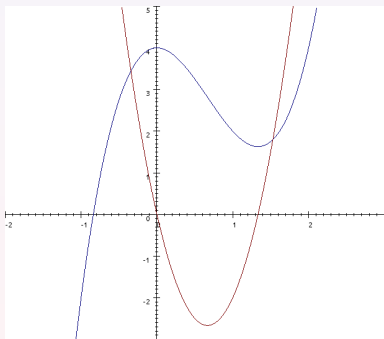
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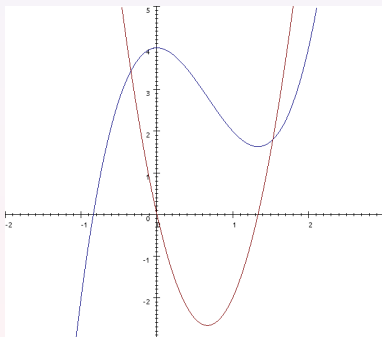
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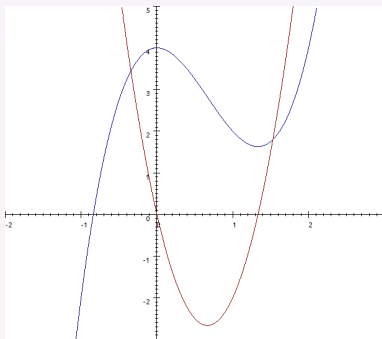
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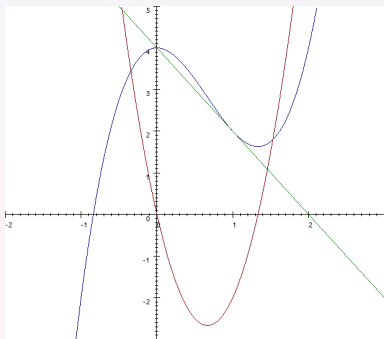
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- For $a=1$,
 $f'(1) = 6(1)^2 - 8(1) = -2$



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Here the velocity varies according to time. Clearly, the derivative is related to this problem, but instantaneous velocity occurs instantly. This problem will again require limits, but instead of tangent lines we will find areas.

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So $x^{11} + C$ represents the *family* of antiderivatives of f .

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Theorems:

Power rule: $\int x^p dx = \frac{x^{p+1}}{p+1} + C$ where $p \neq -1$ is a real number.

Sum rule: $\int (f(x) + g(x))dx = \int f(x)dx + \int g(x)dx$

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- Similarly we can find the indefinite integrals of trigonometric functions; see Table 4.5 pg. 243.

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So $h(t) = t^2 + 2$.

HW: 15, 19, 24, 30, 38, 58

Next time, 5.1: Approximating areas under curves